

MW 1.50 RSE Calculation Report – CS-222 Waffle Slab Check

Project: Structural check of waffle slab (2WS1 & 2WS2) due to Air-cooled magnetic levitation unit (3845 kg)

1. Introduction

Structural check of waffle slab (2WS1 & 2WS2) due to Air-cooled magnetic levitation unit (3845 kg total) at Strafford House Redevelopment, T.P.I.L. 13, 2/F. Complies with MW 1.50 Class I Type A.

2. Structural Synopsis

Waffle slab 750×150 mm thick (C45) + rib beams 200×575 mm, span 9.3 m. One-way action assumed.

3. Design Codes

Hong Kong Concrete Code 2013, CoP Structural Use of Steel 2011.

4. Material Strength

Concrete $f_{cu} = 45$ MPa, $f_y = 460$ MPa. Effective depth $d = 109$ mm.

5. Loading (Step-by-Step)

Plant total = 3845 kg → 3.29 kPa. Concrete plinth SDL = 4.6 kPa. Total Imposed = 7.89 kPa. ULS: 1.4 SDL + 1.6 LL = 14.72 kPa.

6. Structural Analysis (Step-by-Step)

Waffle Slab: $M^* = 3.3$ kNm/m, $V^* = 6.6$ kN. Rib Beam: $M^* = 119.4$ kNm, $V^* = 51.34$ kN.

7. Member Capacity (Step-by-Step)

Slab: $k = 0.006 < 0.156$. $A_{st\ req} = 195$ mm²/m. Provide T12@350 = 323 mm²/m O.K. $v = 0.06 < v_c = 0.709$ O.K.

Rib Beam: $K = 0.047 < 0.156$. Provide 2T20 tension + 2T16 compression. $v = 0.491 < v_c + v_r$. Deflection 17.55 < 27.1 O.K.

8. Deflection Check

Rib beam actual Span/Depth = 17.55 < 27.1 O.K.

9. Conclusion

Waffle slab system is satisfactory. All checks passed per Hong Kong Concrete Code 2013.

10. RSE Sign

RSE Signature: _____ Chan Pak Ming RSE 95/13 Date: 15-Aug-23